

STEP 1.1 · INVESTMENT

Investment Fundamentals

Free cash flows, NPV, IRR, and MIRR — the tools that tell you whether a project is worth your money before you commit a single coin.

START HERE

You're Running the Money for Your Own Company

Read this the way a founder reads it. You are not studying some distant corporation — you are the person who decides where your company's cash goes. Every tool in this step answers a question you will actually face: *is this worth my money, and can I prove it before I commit a single coin?*

This is the first step in Lesson 1 — Investment Appraisal. You'll value projects the way a finance team actually values them: in cash, discounted to today, with a clear rule for when to say yes and when to walk away. Everything in Lesson 2 onwards leans on the machinery you build here, so it's worth slowing down.

Everything that follows is built on this step. NPV, free cash flow, and discounting reappear in every step after it — master them here and the rest of the course is depth, not new ground.

CONCEPT 01

Free Cash Flow – What Your Company Actually Earns

Every investment decision starts with one question: how much cash will this actually produce? Not accounting profit — cash. Profit is what the income statement shows after adjustments for depreciation, accruals, and timing. Cash is what lands in your account.

Free cash flow (FCF) is the cash left after your company has covered its operating costs — salaries, rent, fuel, raw materials, the day-to-day running of the business — paid tax, and reinvested what it needs to keep running and growing. It belongs to everyone who put capital in — both debt holders and shareholders. Think of it as the true earnings of the business.

FCF to the Firm

FCF to the firm is the total cash the business generates for *everyone* who funded it, before any of them are paid. Two groups put money into a company: debt holders (the bank and other lenders) and equity holders (the shareholders who own it). They are not the same — lenders are owed a fixed amount and get paid first, owners take whatever is left. FCF to the firm is the pool both have a claim on. That's why it starts before interest is deducted: interest is the lenders' share, and at this stage you haven't split the pool yet.

Start with operating profit before interest and tax (PBIT) — the operating costs above are already taken out at this point. Add back depreciation — it reduced profit on paper but no cash left when the charge was recorded (it left when you bought the asset). Deduct tax paid, then subtract the cash the business needs to invest to stay operational.

FCF to the Firm

Net operating profit (PBIT)	X
+ Depreciation (non-cash, add back)	X
– Taxation	(X)
Operating cash flows	X
– Replacement of non-current assets (RAI)	(X)
– Incremental non-current assets (IAI)	(X)
– Incremental working capital (IWCI)	(X)
Free cash flow to the firm	X

FCF to Equity

Once you have FCF to the firm, debt holders take their share first. What remains belongs to shareholders.

FCF to Equity

Free cash flow to the firm	X
– Debt interest paid	(X)
– Loan repayments	(X)
+ New debt raised	X
Free cash flow to equity	X

Worked Example — Zambeef Processing Division

Zambeef is reviewing the cash position of one of its cold-chain facilities. Figures are in ZMW '000:

Item	ZMW '000
Operating profit (PBIT)	300
Depreciation	120
Tax paid	(140)
Replacement capex (RAI)	(10)
New investment capex (IAI)	(15)
Increase in working capital (IWC)	(50)
Interest paid	(5)
Loan repaid	(20)

FCF to the firm: $300 + 120 - 140 - 10 - 15 - 50 = \text{ZMW } 205,000$

FCF to equity: $205 - 5 - 20 = \text{ZMW } 180,000$

Free cash flow is what remains after the company has kept running, replaced its assets, and funded its growth — it is what actually belongs to investors, not the profit number on the income statement.

CONCEPT 02

Net Present Value – The Right Way to Say Yes or No

Your company is deciding whether to invest ZMW 24,000 in a new distribution route. The route will generate cash for five years. The question is not whether total cash in exceeds ZMW 24,000 — it's whether those future cash flows are worth ZMW 24,000 today.

Money received in the future is worth less than money today. ZMW 7,800 arriving in one year is not the same as ZMW 7,800 now — because ZMW 7,800 now could be invested and earn a return. NPV accounts for this by discounting all future cash flows back to today's value.

Discount Factor = $(1 + r)^{-n} = 1 / (1 + r)^n$

At cost of capital 10%:

Year 1: $1 / (1.10)^1 = 0.909$

Year 2: $1 / (1.10)^2 = 0.826$

Year 3: $1 / (1.10)^3 = 0.751$

Rule: NPV > 0 → accept. NPV < 0 → reject.

Worked Example — Distribution Project

Year	Cash Flow (ZMW)	Discount Factor (10%)	Present Value (ZMW)
0	(24,000)	1.000	(24,000)
1	7,800	0.909	7,090
2	6,000	0.826	4,956
3	4,200	0.751	3,154
4	7,400	0.683	5,054
5	9,200	0.621	5,713
NPV =			1,967

NPV = ZMW 1,967. The project creates ZMW 1,967 of value in today's money, above and beyond recovering the investment and earning the required 10% return. Accept it.

A positive NPV means today's value of future earnings exceeds the cost of the investment — the project creates wealth for your company.

Your company is comparing two projects. Project A: NPV of ZMW 1,967, requires ZMW 24,000. Project B: NPV of ZMW 1,400, requires ZMW 10,000. You can only fund one. Which do you choose — and does the answer change if you have ZMW 50,000 available? What does your decision reveal about how you think about allocating capital inside the business?

CONCEPT 03

Discounted Payback – When Do You Get Your Money Back?

Discounted payback period (DPP) is how long it takes for the cumulative present value of cash inflows to recover the initial investment. It's a stricter version of simple payback — you're asking when you get your money back in real terms, after accounting for the time value of money.

A project is acceptable if its DPP falls within the company's target. Using the same ZMW 24,000 distribution project:

Year	Cash Flow (ZMW)	Discount Factor	Present Value (ZMW)	Cumulative PV (ZMW)
0	(24,000)	1.000	(24,000)	(24,000)
1	7,800	0.909	7,090	(16,910)
2	6,000	0.826	4,956	(11,954)
3	4,200	0.751	3,154	(8,800)
4	7,400	0.683	5,054	(3,746)
5	9,200	0.621	5,713	1,967

DPP = 4 + (3,746 / 5,713) = 4 + 0.66 = 4.66 years

The remaining shortfall after Year 4 is ZMW 3,746.

Year 5 brings in PV of ZMW 5,713 — recovery happens 3,746/5,713 through Year 5.

If the target payback is 4 years, this project does not meet it.

Note: the project still has a positive NPV. DPP and NPV can conflict.

Discounted payback tells you when you recover your investment in real terms — use it as a liquidity check, not as the main decision rule. A project can miss the payback target and still be worth doing.

CONCEPT 04

Internal Rate of Return – The Break-Even Discount Rate

The internal rate of return (IRR) is the discount rate that makes NPV equal to zero. It's the rate your project earns on the capital tied up in it. If your company's cost of capital is 10% and the project earns 13%, it clears the hurdle. Accept it.

IRR is found by interpolation: calculate NPV at two different discount rates — one that gives a positive NPV and one that gives a negative NPV — then estimate where NPV crosses zero.

$$\text{IRR} = A + [a / (a - b)] \times (B - A)$$

A = lower discount rate with positive NPV (a)

B = higher discount rate with negative NPV (b)

Worked Example

Same ZMW 24,000 project. Calculate NPV at 10% and 14%:

Year	Cash Flow	DF at 10%	PV at 10%	DF at 14%	PV at 14%
0	(24,000)	1.000	(24,000)	1.000	(24,000)
1	7,800	0.909	7,090	0.877	6,841
2	6,000	0.826	4,956	0.769	4,614
3	4,200	0.751	3,154	0.675	2,835
4	7,400	0.683	5,054	0.592	4,381
5	9,200	0.621	5,713	0.477	4,775
		NPV =	1,967	NPV =	(554)

$$\text{IRR} = 10\% + [1,967 / (1,967 + 554)] \times (14\% - 10\%)$$

$$= 10\% + [1,967 / 2,521] \times 4\%$$

$$= 10\% + 3.12\% = 13.12\%$$

IRR (13.12%) > cost of capital (10%) → accept.

Where IRR Fails

IRR has two problems that matter in practice.

- **Multiple IRRs.** If a project has cash flows that change sign more than once — an outflow mid-project, for example — there can be two or more IRRs. The accept/reject rule breaks down completely.
- **Unrealistic reinvestment assumption.** IRR assumes that every kwacha of cash generated by the project is immediately reinvested at the IRR itself. For a project with 13% IRR, that means finding 13% returns for reinvested cash throughout the project life. That is almost never possible. NPV avoids this problem by assuming reinvestment at the cost of capital.

IRR tells you the break-even discount rate — not the actual return — unless cash flows can genuinely be reinvested at that same rate throughout the project. In most real projects, they cannot.

CONCEPT 05

Modified IRR – A More Honest Return

MIRR fixes IRR's reinvestment problem by separating the two rates: your cost of capital for outflows, and a realistic reinvestment rate for inflows. It asks: given what you can actually earn on cash coming back during the project, what is the real return?

$$\text{MIRR} = \sqrt[n]{\text{FVCF} / \text{PVCF}} - 1$$

FVCF = future value of inflows, compounded at the reinvestment rate

PVCF = present value of outflows, discounted at cost of capital

n = project life in years

Step 1: Discount all outflows to Year 0 at cost of capital.

Step 2: Compound all inflows to Year n at the reinvestment rate.

Step 3: Calculate the rate that equates PVCF and FVCF over n years.

Worked Example

Same ZMW 24,000 project. Cost of capital: 10%. Reinvestment rate: 12%.

Step 1: Outflow at Year 0 = ZMW 24,000 (already at present value).

Step 2: Compound each inflow to Year 5 at 12%:

Year	Cash Flow (ZMW)	Compound Factor	Future Value at Year 5 (ZMW)
1	7,800	$1.12^4 = 1.5735$	12,273
2	6,000	$1.12^3 = 1.4049$	8,430
3	4,200	$1.12^2 = 1.2544$	5,268
4	7,400	$1.12^1 = 1.1200$	8,288
5	9,200	$1.12^0 = 1.0000$	9,200
Total FVCF =			43,459

$$\text{MIRR} = \sqrt[5]{(43,459 / 24,000)} - 1$$

$$= \sqrt[5]{1.8108} - 1$$

$$= 1.1260 - 1$$

$$= 12.6\%$$

MIRR (12.6%) > cost of capital (10%) → project is acceptable.

Compare: IRR was 13.12%. MIRR gives the lower, more honest figure.

IRR overstated the return by assuming 13.12% reinvestment.

MIRR replaces IRR's impossible reinvestment assumption with a rate you actually expect to earn — giving a more accurate picture of what the project truly returns to your business.

CONCEPT 06

Inflation and Tax – Two Adjustments You Cannot Skip

Handling Inflation

Inflation affects both cash flows and discount rates. You can handle it two ways — both give the same NPV when done correctly.

- **Nominal method.** Inflate each cash flow at its specific rate, then discount at the nominal cost of capital. Use this when different items inflate at different rates — wages at 8%, raw materials at 5%, selling prices at 6%.
- **Real method.** Strip inflation from cash flows and discount at the real rate. Only works when everything inflates uniformly at the general rate.

Fisher Equation: $(1 + \text{nominal rate}) = (1 + \text{real rate}) \times (1 + \text{inflation rate})$

Example: real rate 5%, inflation 8%

Nominal rate $= (1.05 \times 1.08) - 1 = 13.4\%$

Handling Tax

Tax affects investment appraisal in two ways. First, the company pays tax on operating cash flows — this is an outflow that reduces FCF. Second, tax allowable depreciation (capital allowances) reduces taxable profit, generating a tax saving that acts as a cash inflow.

Always include: (1) tax on operating cash flows as an outflow, and (2) the tax saving from capital allowances as an inflow. Leaving either out will give you the wrong NPV.

Tax on operating flows reduces your FCF. Capital allowances reduce your tax bill. Both are real cash movements and both change the NPV of a project.

First Quantum is evaluating a ZMW 3 million processing upgrade. Cash flows: ZMW 600,000 in Year 1, increasing by ZMW 160,000 each year for five more years. The cost of capital is 21% and the reinvestment rate is 8%. Would you calculate IRR or MIRR first — and why does the gap between those two rates matter more for a long-horizon project than a short one?

REFERENCE

Key Terms

Term	Definition
Free Cash Flow (FCF)	Cash available after operating costs, tax, and reinvestment in working capital and fixed assets
FCF to the Firm	FCF available to all providers of capital — debt holders and equity holders
FCF to Equity	FCF remaining for shareholders after interest and loan repayments
PBIT	Profit before interest and tax — the starting point for FCF calculation
Net Present Value (NPV)	Sum of all discounted future cash flows minus the initial investment
Discount Factor	Multiplier that converts a future cash flow to present value: $1 / (1+r)^n$
Discounted Payback Period (DPP)	Time to recover the initial investment using present-value cash flows
Internal Rate of Return (IRR)	The discount rate at which NPV = 0; the project's break-even rate
Modified IRR (MIRR)	IRR recalculated using a realistic reinvestment rate, giving a more accurate return figure
Reinvestment Rate	The rate at which interim cash flows are actually expected to be reinvested; used in MIRR
Cost of Capital	The required return used as the discount rate; derived from WACC/CAPM in Step 2.1
Nominal Rate	The stated rate, including inflation
Real Rate	The rate adjusted for inflation
Fisher Equation	$(1 + \text{nominal}) = (1 + \text{real}) \times (1 + \text{inflation})$ — links the three rates
Capital Allowance	Tax-deductible depreciation that generates a tax saving on investments
RAI / IAI / IWCi	Replacement assets / Incremental assets / Incremental working capital — the three investment deductions in FCF

OUTCOMES

What You Should Now Be Able to Do

1. Calculate free cash flow to the firm and free cash flow to equity from accounting data
2. Apply the NPV decision rule and explain in plain terms what a positive NPV means

3. Calculate the discounted payback period and identify when it conflicts with NPV
4. Compute IRR by linear interpolation and apply the accept/reject rule
5. Explain the two main limitations of IRR: multiple IRRs and the reinvestment assumption
6. Calculate MIRR and explain why it gives a more accurate return figure than IRR
7. Handle inflation in NPV analysis using both the nominal and real methods
8. Identify the two ways tax affects appraisal: tax on operating flows and capital allowance savings

Where You Go Next

You came in with the full map of the course (look back at the first page). You now own the first piece of it — the toolkit every other step leans on. **Step 1.2 — Advanced Investment Appraisal** takes the same NPV machinery and handles the harder cases: projects financed partly by debt (APV), and what you do when you can't fund every good project at once (capital rationing).

Nothing in Step 1.2 replaces what you learned here — it extends it. That is the whole idea: you are not starting over each step, you are deepening one picture you already hold.

This is Step 1.1 in the Corporate Finance series running inside the Bookless study group on Slack. The channel for this topic is **#cf-investment** — that's where BAC4301 students are working through investment appraisal together, sharing past paper questions, and picking apart the calculations from this step.

If you're already there, you know where to find it. If not, [join the group here](#).